



Whale Optimization Algorithm (WOA)

Course: Planning Techniques

Department : Artificial Intelligence

Prepared by:

- Raghad Gmal Abdelmoez
- Rawaa Ehab Elsayed
- Sama Essam Abdelfatah
- Shahd Ibrahim Mahrous
- Tasneem Hussien Ashry

Supervised by :

- DR. Mahmoud Abdelaal
- ENG. Youssef Elbaroudy

Faculty of Computers & Artificial Intelligence
Benha University

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Whale Optimization Algorithm (WOA)



Abstract:

The Whale Optimization Algorithm (WOA) is a nature-inspired meta-heuristic that mathematically emulates the bubble-net hunting strategy of humpback whales (*Megaptera novaeangliae*). Unlike swarm intelligence algorithms that replicate coordinated group behaviors, WOA derives its mechanisms from the solitary foraging tactics of individual humpback whales. Central to this is the spiral bubble-net maneuver—a sophisticated predatory behavior in which a whale generates a contracting circle of bubbles while ascending in a spiral trajectory to concentrate and capture prey.. This biological phenomenon translates into a dual-phase optimization framework comprising: (1) an exploitation phase characterized by shrinking encircling mechanisms and logarithmic spiral position updates, and (2) an exploration phase governed by stochastic search agents operating independently of collective information sharing.

Since its introduction by Mirjalili and Lewis in 2016, WOA has garnered substantial scholarly attention across diverse engineering domains. The algorithm's conceptual simplicity, minimal parameter requirements, rapid convergence characteristics, and effective balance between diversification and intensification have established it as a competitive alternative to classical methods such as Particle Swarm Optimization (PSO), Genetic Algorithms (GA), and Grey Wolf Optimizer (GWO). Benchmark evaluations demonstrate that WOA achieves superior solution accuracy and convergence behavior across multimodal, high-dimensional, and constrained optimization landscapes.

Contemporary research has produced numerous algorithmic variants addressing WOA's inherent limitations, including sensitivity to initial population conditions, susceptibility to local optima entrapment, and convergence deceleration in complex search spaces. Notable improvements incorporate chaotic mapping mechanisms for population initialization (e.g., PWLCM, Tent maps) , adaptive inertia weighting strategies, and hybridization with differential evolution, simulated annealing, and other metaheuristics. The multi-objective extension (MOWOA) has emerged as particularly significant, with bibliometric analyses revealing concentrated research activity in computer science and engineering disciplines, where China (36.9%) and India (18.5%) represent the largest contributing nations.

The algorithm's practical efficacy extends to feature selection, neural network optimization, cloud computing resource allocation, and engineering design problems, with hybrid implementations demonstrating robust performance across diverse application domains. This review synthesizes the biological foundations, algorithmic architecture, performance characteristics, and contemporary developments of WOA, providing a comprehensive resource for researchers seeking to apply or extend this versatile optimization methodology.



Chapter 1: Introduction

1.1 Introduction

Optimization, the process of finding the best possible solution from a set of feasible alternatives, stands as a cornerstone of scientific inquiry, engineering design, and operational decision-making. From minimizing energy consumption in large-scale networks to maximizing the efficiency of complex industrial processes, the pursuit of optimality is a pervasive challenge. Traditional optimization techniques, such as gradient-based methods, linear programming, and dynamic programming, have proven effective for problems characterized by convexity, differentiability, and moderate dimensionality. However, the contemporary landscape is increasingly dominated by complex, real-world problems that defy these classical assumptions. These problems are often non-linear, non-convex, multimodal, and high-dimensional, with discontinuous or noisy objective functions and intricate constraint landscapes. In such contexts, conventional deterministic algorithms frequently fail to converge a global optimum, becoming trapped in suboptimal local solutions or exhibiting prohibitive computational costs.

This paradigm has catalyzed the development and widespread adoption of meta-heuristic algorithms. Unlike deterministic methods, meta-heuristics are high-level, problem-independent frameworks that guide subordinate heuristics to explore a solution space efficiently. They are inspired by natural phenomena, leveraging stochastic elements, and iterative refinement to balance exploration (searching new regions) and exploitation (refining known good regions). This class of algorithms, which includes genetic algorithms (GAs), particle swarm optimization (PSO), ant colony optimization (ACO), and simulated annealing (SA), has demonstrated remarkable success in addressing a vast array of challenging optimization problems across diverse disciplines.

Among the more recent additions to this family is the Whale Optimization Algorithm (WOA), a nature-inspired meta-heuristic introduced by Seyedali Mirjalili and Andrew Lewis in 2016. WOA distinguishes itself through its elegant emulation of the unique hunting behavior of humpback whales, specifically a foraging strategy known as bubble-net feeding. The algorithm mathematically models this behavior through three primary phases: encircling prey, bubble-net attacking (exploitation), and searching for prey (exploration), with a seamless transition between these phases orchestrated by dynamically adjusted parameters. This balance between exploration and exploitation allows the algorithm to efficiently navigate complex search landscapes. WOA has rapidly gained prominence due to its simplicity, few controlling parameters, and impressive performance in terms of convergence speed and solution accuracy when benchmarked against established algorithms.



1.2 Research Objectives and Scope

The primary objective of this research is to conduct a comprehensive exploration of the Whale Optimization Algorithm, evaluating its efficacy as a robust optimization tool for complex problems where traditional methods are inadequate. This study aims to achieve several key goals:

Assess Performance: To rigorously evaluate WOA's solution accuracy, convergence behavior, and ability to avoid local optima across a spectrum of benchmark functions and real-world engineering problems, particularly those characterized by non-linearity, non-convexity, and high dimensionality.

Investigate Applicability: To demonstrate the algorithm's adaptability by examining its performance across different problem dimensions and its integration with constraint-handling techniques, thereby extending its utility to constrained optimization tasks.

Explore Enhancements: To critically analyze the foundational algorithm's limitations—such as its potential for premature convergence—and investigate the potential for further improvements through modifications or hybridization with other optimization strategies.

The scope of this work encompasses the application of WOA to continuous mathematical optimization problems, and an analysis of its scalability in high-dimensional contexts. By achieving these objectives, this research seeks to contribute to the growing body of knowledge on nature-inspired algorithms and provide practical insights for leveraging WOA in challenging optimization scenarios.



Chapter 2: Literature Review

2.1 Introduction

In recent years, nature-inspired optimization techniques have attracted growing attention because of their ability to solve complex problems that traditional mathematical methods struggle to handle. One of the most notable algorithms in this category is the Whale Optimization Algorithm (WOA), proposed by Mirjalili and Lewis in 2016. The algorithm simulates the hunting behavior of humpback whales, particularly the bubble-net feeding strategy, to search for optimal solutions.

Since its development, WOA has been applied to a wide range of scientific and engineering problems. Researchers have explored its use in areas such as energy systems, robotics, machine learning, image processing, and scheduling. The increasing number of studies published on WOA reflects its effectiveness and adaptability to different types of optimization challenges.

2.2 Power and Energy Systems

Electrical power systems require continuous optimization to ensure reliable operation while minimizing costs and losses. One common problem is economic load dispatch, where electricity must be generated at the lowest possible cost while still meeting demand and operational constraints. Traditional optimization techniques often struggle when fuel cost functions are nonlinear or when system constraints are complex. WOA has shown the ability to find cost-effective solutions in such cases by exploring multiple candidate solutions simultaneously.

Another important application is optimal power flow, which aims to determine the best operating conditions for a power network. This includes adjusting generator outputs, transformer settings, and other control variables to minimize losses and maintain voltage stability. WOA can efficiently search through a large number of possible configurations to identify suitable operating points.

In distribution networks, capacitor placement plays a key role in improving energy efficiency. Installing capacitors at optimal locations reduces reactive power losses and enhances voltage profiles. WOA can determine both the size and location of capacitors while considering system constraints, leading to improved performance and reduced operational costs.



2.3 Robotics and Autonomous Systems

Autonomous robots must operate in environments that are often unpredictable and dynamic. Path planning is a critical task that involves determining a safe and efficient route from a starting position to a destination. WOA has been successfully used to generate collision-free paths while minimizing travel distance or time. The algorithm evaluates multiple possible routes and gradually improves them, increasing the likelihood of finding a globally efficient solution.

Trajectory optimization is another important problem in robotics. Robots and unmanned vehicles need smooth motion patterns to reduce energy consumption and mechanical stress. By optimizing control points along a trajectory, WOA helps produce movements that are both efficient and accurate.

In addition, WOA has been applied to multi-robot coordination, where several robots must work together without interfering with each other. Efficient coordination requires solving complex optimization problems involving task allocation and motion planning simultaneously. The algorithm's population-based nature makes it suitable for exploring such large solution spaces.

2.4 Machine Learning and Data Analysis

Optimization is a fundamental component of machine learning. Many models depend on selecting appropriate parameters to achieve good predictive performance. WOA has been widely used to tune hyperparameters of algorithms such as neural networks and support vector machines. By searching for parameter combinations that minimize prediction error, the algorithm can improve model accuracy without requiring exhaustive manual testing.

Feature selection is another area where WOA has been applied. High-dimensional datasets often contain redundant or irrelevant attributes that increase computational cost and may reduce model performance. WOA can identify subsets of features that preserve essential information while removing unnecessary variables. This results in simpler models that train faster and generalize better new data.

Researchers have also used WOA for clustering tasks, where the objective is to group similar data points together. The algorithm helps determine cluster centers that minimize intra-group distance while maximizing separation between clusters. Such techniques are useful in fields ranging from market analysis to bioinformatics.



2.5 Image Processing and Computer Vision

Digital images contain large amounts of data, and many processing tasks require optimization of parameters. WOA has been applied to image segmentation, which involves dividing an image into meaningful regions. Accurate segmentation is essential in applications such as medical diagnosis, object recognition, and remote sensing.

In medical imaging, for example, segmentation algorithms are used to isolate organs or abnormal tissues from scans. WOA can optimize threshold values or region boundaries, improving the accuracy of automated analysis systems. This reduces the need for manual intervention by medical experts.

Another application is image enhancement. Techniques such as contrast adjustment and noise reduction depend on selecting appropriate parameters. WOA can automatically determine values that improve visual quality while preserving important details. These improvements are useful in surveillance, satellite imaging, and industrial inspection.

2.6 Scheduling and Resource Allocation

Scheduling problems occur in many practical situations, including manufacturing, transportation, and computing systems. These problems typically involve assigning tasks to limited resources while optimizing performance measures such as completion time or cost.

In job-shop scheduling, multiple jobs must be processed on several machines with specific processing sequences. Finding the optimal schedule is computationally difficult because the number of possible arrangements grows rapidly with problem size. WOA can explore different schedules and identify those that minimize overall production time.

Cloud computing environments also require efficient resource allocation. Service providers must assign virtual machines and computing resources to users while maintaining performance and minimizing energy consumption. WOA helps balance workloads across servers, reducing delays and preventing system overload.

2.7 Communication Networks and IoT Systems

Modern communication systems involve complex decisions related to routing, bandwidth allocation, and energy management. WOA has been applied to optimize these aspects, particularly in wireless sensor networks where devices operate with limited battery power.



Whale Optimization Algorithm (WOA)



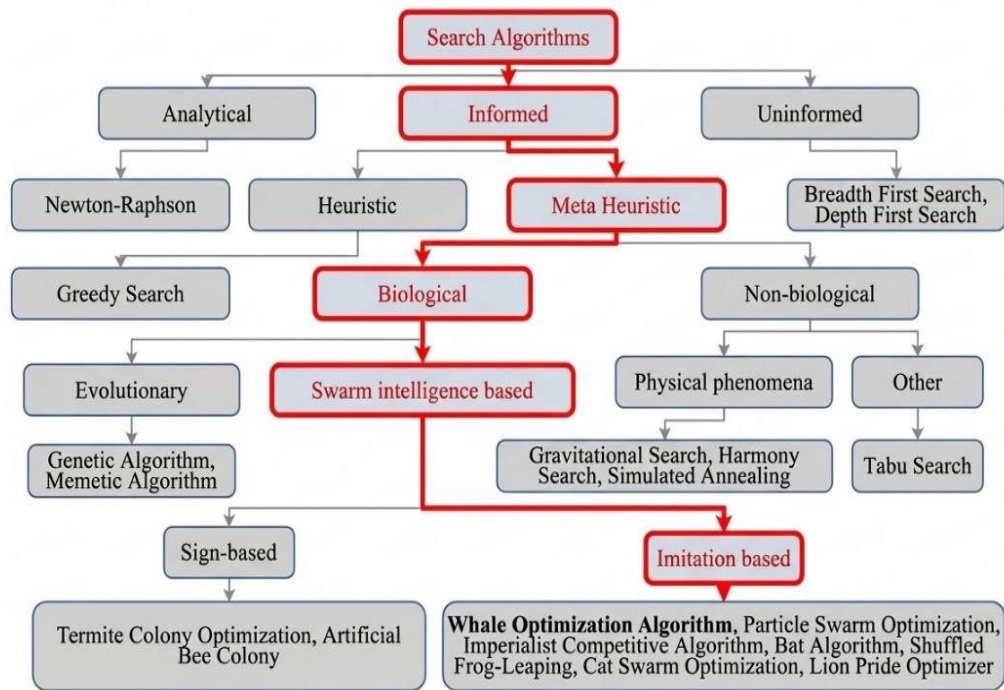
One important task is determining optimal sensor placement to ensure adequate coverage of a monitored area. WOA can search for configurations that maximize coverage while minimizing the number of sensors used. This reduces deployment costs and energy consumption.

Routing optimization is another critical issue. Data packets must travel through the network efficiently while avoiding congestion and minimizing delays. WOA helps identify routing paths that balance these requirements.

In Internet of Things environments, thousands of interconnected devices generate and transmit data continuously. Efficient management of communication and resources is essential for maintaining system reliability. WOA provides a flexible approach for adapting to changing conditions in large-scale distributed systems.

Chapter 3: The proposed algorithm

3.1 Algorithm Classification



3

1. Informed Search Algorithm

WOA is considered an informed search algorithm because it uses information about the best solution found so far (X^*) to guide the movement of other solutions. This allows the algorithm to focus on promising regions in the search space and improves convergence speed.

2. Meta-Heuristic Algorithm

WOA belongs to the class of meta-heuristic algorithms, which are general-purpose optimization methods designed to solve complex problems without requiring specific mathematical models. These algorithms are flexible and can be applied to a wide range of optimization problems.



3. Bio-inspired (Swarm Intelligence) Algorithm

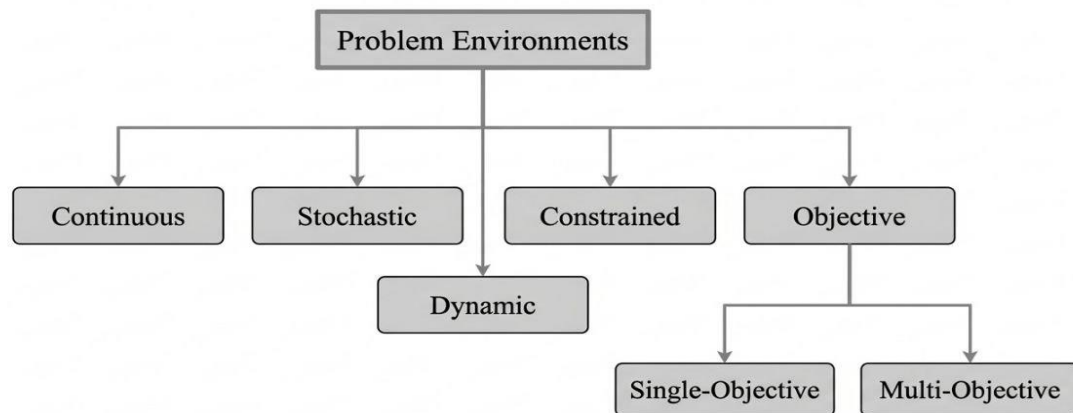
WOA is a bio-inspired algorithm based on the social behavior of humpback whales. It mimics the bubble-net feeding strategy, where a group of whales cooperates to capture prey. This makes it part of swarm intelligence algorithms, where multiple agents interact to find optimal solutions.

4. Imitation-Based Algorithm

The Whale Optimization Algorithm (WOA) is considered an imitation-based algorithm as it relies on following the best solution found so far (X^*) to guide the movement of whales within the search space. This behavior allows the algorithm to exploit promising regions effectively and accelerates convergence toward the optimal solution.

3.2 Problem Environment Classification

The Whale Optimization Algorithm (WOA) can be applied in different problem environments. These environments define the characteristics of the optimization problem and how the algorithm searches for the optimal solution.



1. Continuous Environment

In continuous environments, all decision variables can take real values within a defined range. This allows whales to move smoothly across the search space, exploring intermediate solutions and refining candidate solutions with fine granularity. Continuous optimization is the standard setting for WOA and is widely used in engineering design, parameter tuning, and numerical



optimization tasks. The ability to traverse continuous domains ensures that WOA can accurately approximate optimal solutions and adaptively adjust movements based on the fitness landscape.

2. Stochastic Environment

WOA operates effectively in stochastic environments, where randomness is inherent in the search process. Random variables such as rrr , ppp , and lll control the movement of whales, introducing diversity in exploration and preventing premature convergence to local optima. Stochastic behavior allows the algorithm to investigate multiple regions of the search space simultaneously, enhancing the robustness of the optimization. This characteristic is particularly valuable in dynamic or uncertain problem settings, where deterministic algorithms may fail to identify the global optimum due to limited exploration.

3. Constrained Environment

Many real-world optimization problems involve constraints on the variables, such as boundaries, resource limitations, or specific functional requirements. WOA can handle such constrained problems through appropriate constraint-handling techniques, ensuring that solutions remain feasible throughout the search process. By incorporating constraints into the fitness evaluation or by modifying the position update rules, WOA can navigate restricted search spaces without violating problem requirements, making it suitable for practical applications in engineering, scheduling, and resource allocation.

4. Objective Environment

WOA is capable of addressing both single-objective and multi-objective optimization problems. In single-objective optimization, the algorithm focuses on optimizing a single fitness function, aiming to locate the global optimum efficiently. In multi-objective scenarios, WOA can be adapted to optimize multiple conflicting objectives simultaneously, balancing trade-offs between competing criteria. This flexibility allows WOA to be applied to complex engineering problems, financial modeling, and decision-making tasks that require simultaneous consideration of multiple performance measures.

5. Dynamic Environment

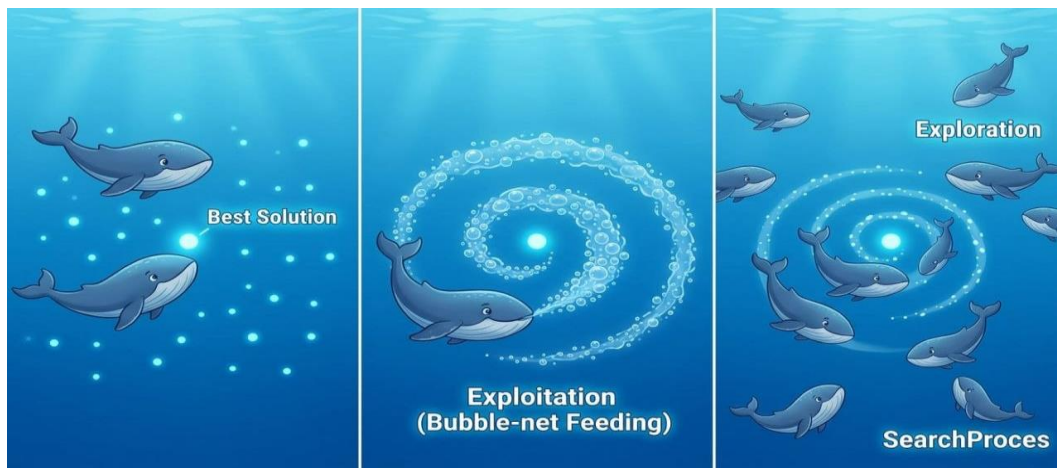
Dynamic environments are characterized by changes in the problem structure or fitness landscape over time. For example, the optimal solution may shift due to varying constraints, evolving objectives, or external disturbances. WOA can adapt to such dynamic conditions by updating the positions of whales and recalculating the best-known solutions at each iteration. This adaptability ensures that the algorithm remains effective in real-time applications, such as

network routing, robotics, and adaptive control systems, where the problem continuously evolves.

3.3 Key Characteristics of WOA

1. Balance between Exploration and Exploitation

WOA maintains a balance between global search (exploration) and local search (exploitation) through its encircling and spiral bubble-net behavior. Exploration allows the algorithm to investigate new regions of the search space, while exploitation enables the refinement of promising solutions. This dynamic balance is controlled by the algorithm's parameters, ensuring that the population neither converges prematurely nor fails to exploit high-quality solutions effectively.



2. Population-Based Search

The algorithm employs a population of whales that move simultaneously in the search space. Each whale represents a candidate solution, and the collective movement enhances diversity, robustness, and the probability of locating the global optimum. This multi-agent approach is a key factor in preventing the algorithm from stagnating in local optima.

3. Bio-Inspired Strategy

WOA is inspired by the natural bubble-net feeding strategy of humpback whales. The algorithm mimics the social and cooperative behavior of whales in capturing prey. This bio-inspired mechanism enables the algorithm to perform coordinated search, where agents influence each



other while collectively moving toward the best-known solution. Such swarm intelligence contributes to improved convergence rates and search efficiency.

4. Stochastic Elements

The use of random variables (r, p, l) introduces controlled stochasticity in the movement of whales. This randomness allows WOA to explore diverse regions of the search space, preventing premature convergence to sub-optimal solutions. The stochastic nature of the algorithm enhances its robustness and adaptability in dynamic and uncertain environments.

5. Flexible Adaptation

WOA demonstrates high flexibility and can be adapted to a wide range of optimization problems. It is effective in continuous, discrete, constrained, unconstrained, single-objective, and multi-objective optimization scenarios. Its flexible framework allows researchers to modify position update rules, control parameters, or fitness evaluation strategies to suit specific problem requirements.

3.4 Steps of Algorithm

Step 1: Searching for Prey

In Nature:

Humpback whales search for prey randomly in the ocean using their sensory abilities.

In Algorithm:

The algorithm initializes a population of whales randomly in the search space. Each whale represents a candidate solution.

Step 2: Identifying the Best Prey

In Nature:

Whales detect the position of prey once it is found.

In Algorithm:

The best solution is identified based on the fitness function. This solution is considered the target prey (X^*).

Step 3: Encircling the Prey



Whale Optimization Algorithm (WOA)



In Nature:

Whales swim around the prey to trap it.

In Algorithm:

Other solutions update their positions toward the best solution using mathematical equations that simulate encircling behavior.

Step 4: Bubble-Net Feeding (Spiral Movement)

In Nature:

Whales create spiral bubbles and move in a helix-shaped path to capture prey.

In Algorithm:

A spiral equation is used to update the position of solutions around the best solution, simulating this behavior.

Step 5: Random Search for Prey

In Nature:

If no prey is detected, whales continue searching randomly.

In Algorithm:

The algorithm performs exploration by selecting random solutions and moving toward them instead of the best solution.

Step 6: Attacking the Prey

In Nature:

Whales finally attack and capture the prey.

In Algorithm:

The algorithm converges toward the optimal solution after several iterations.

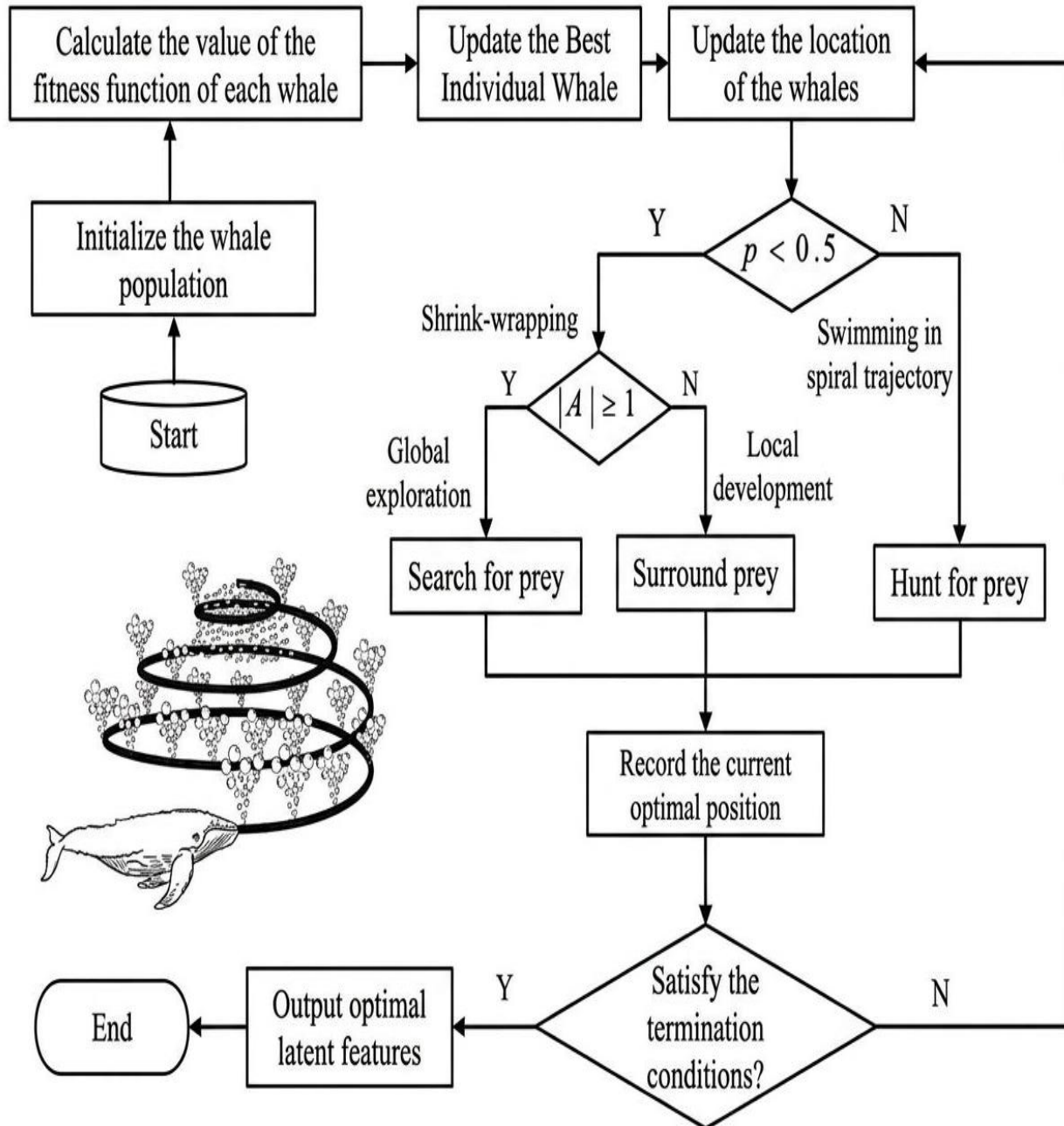


Whale Optimization Algorithm (WOA)



Behavior	Nature	Algorithm
Searching	Random movement	Initialize population randomly
Encircling	Swim around prey	Move toward best solution X^*
Spiral attack	Helix movement	Update position with spiral equation
Exploration	Search new areas	Random movement based on A, C
Attack	Capture prey	Convergence to optimal solution

3.5 Flowchart





3.6 Pseudocode & Explanation

3.6.1 Pseudocode

1. Initialize whale population $X_i (i = 1, \dots, N)$
2. Define fitness function $f(X)$
3. Calculate fitness of each whale
4. $X^* = \text{best solution}$
5. For $t = 1$ to max_iterations :
 6. For each whale X_i
 7. Compute A, C, p, l
 8. If $p < 0.5$:
 9. If $|A| < 1$:
 10. $X_i(t + 1) = X^* - A \cdot |C \cdot X^* - X_i|$
//Encircling
 11. Else:
 12. $X_{rand} = \text{randomwhale}$
 - 13.
 14. Else:
 - 15.
16. Update fitness
17. Update X^* if better solution found
18. End For
19. Return X^*



3.6.2 Pseudocode Explanation

The pseudocode describes the main steps of the Whale Optimization Algorithm (WOA) in a structured way. It explains how the algorithm initializes solutions, updates their positions, and gradually converges toward the optimal solution.

First, the algorithm starts by initializing a population of whales randomly within the search space. Each whale represents a candidate solution to the optimization problem. After initialization, a fitness function is used to evaluate the quality of each solution, and the best solution is identified.

Next, the algorithm enters an iterative process. In each iteration, every whale updates its position based on certain behavioral rules inspired by humpback whales.

A set of control parameters is calculated to guide the movement of each whale. These parameters determine whether the whale will exploit the current best solution or explore new areas in the search space.

If a randomly generated probability value is less than a threshold, the whale follows one of two strategies. If the control parameter indicates exploitation, the whale moves toward the best solution found so far. Otherwise, it explores the search space by moving toward a randomly selected whale.

On the other hand, if the probability value is greater than the threshold, the whale performs a spiral movement around the best solution. This simulates the bubble-net feeding behavior of humpback whales.

After updating all whale positions, the fitness of the new solutions is evaluated again. If a better solution is found, it replaces the previous best solution.

This process continues for a fixed number of iterations or until a stopping condition is met. Finally, the algorithm returns the best solution obtained.

3.7 Control Parameters and Their Roles

In the Whale Optimization Algorithm (WOA), a set of control parameters is used to regulate the movement of whales within the search space. These parameters are essential for balancing **exploration** (searching for new regions) and **exploitation** (refining the current best solution).



1. Parameter a

The parameter a decreases linearly from 2 to 0 over the course of iterations. It is mainly used to calculate coefficient A , which directly influences the movement behavior of whales.

- Large values of encourage **exploration**.
- Small values of encourage **exploitation**.

2. Parameter a'

The parameter a' is a linearly decreasing value similar to a , but it is specifically used in calculating the spiral shape parameter b . It helps control the contraction and flexibility of the spiral movement during iterations.

3. Parameter A

The coefficient A determines whether the whale performs exploration or exploitation.

- If $|A| < 1$ the whale moves toward the current best solution (**exploitation**).
- If $|A| \geq 1$ the whale explores new regions of the search space (**exploration**).

Thus, A is the primary factor controlling the transition between global and local

Search.

4. Parameter C

The coefficient C is used to compute the distance between the whale and its target (either the best solution or a randomly selected solution). It introduces randomness into the movement, which helps diversify the search process.

5. Parameter p

The parameter p is a random number in the range $[0,1]$. It determines the movement strategy of the whale:

- If $p < 0.5$ the whale performs encircling or exploration behavior.
- If $p \geq 0.5$ the whale executes a spiral movement around the best solution.

6. Parameter b

In this implementation, the parameter b is dynamically calculated using a random value and a'



This allows the spiral shape to vary during iterations, enhancing the diversity of the search and preventing premature convergence.

7. Parameter l

The parameter l is a random number in the range $[-1,1]$. It controls the shape and direction of the spiral movement during the bubble-net feeding behavior.

8. Parameter r

The parameter r is a random number in the range $[0,1]$. It is used to introduce stochastic behavior in the algorithm and to compute the coefficient vectors A and C, helping balance exploration and exploitation.

3.8 Mathematical Model

3.8.1 Parameter Update

At each iteration, the control parameters are updated to regulate the behavior of the algorithm.

The parameter a decreases linearly over iterations according to : $a = 2 - \left(\frac{2t}{\text{Max Iteration}} \right)$

Additionally, a secondary parameter a' is defined as: $a' = -1 + \left(\frac{-t}{\text{Max Iteration}} \right)$

The coefficient vector A is calculated as: $A = 2 \cdot a \cdot r - a$

and the coefficient vector C is given by: $C = 2 \cdot r$

The parameter b is dynamically calculated as: $b = r \cdot (a' - L) + L$

Where L is a random number in the range $[0,1]$.

Another random parameter $p \in [0,1]$ is used to determine the movement strategy.

3.8.2 Position Updating Mechanism

The position of each whale is updated based on the value of p and A .

When: $p < 0.5$ and $|A| < 1$

The algorithm performs **encircling behavior**, where whales move toward the current best solution.



3.8.3 Encircling Prey Equation

The distance between the current whale and the best solution is calculated as: $D = |C \cdot X^* - X|$

The position is then updated using: $X(t + 1) = X^* - A \cdot D$

This equation enables the whale to move closer to the best solution, enhancing the exploitation process.

3.8.4 Boundary Handling

After updating positions, some values may fall outside the allowed search range $[0,1]$. In such cases, the values are adjusted to remain within the valid boundaries. For example, negative values are set to zero.

3.8.5 Fitness Evaluation and Best Solution Update

After updating all whale positions, the fitness of each solution is recalculated using the objective function. The best solution X^* is then updated if a better solution is found.

This iterative process allows the algorithm to gradually converge toward the optimal solution by continuously improving the population of candidate solutions.



Chapter 4: Numerical Example

The problem is to minimize the following function:

$$F(x) = 2.47 X_1 + 7.35 X_2$$

While X_1 and X_2 $[0,1]$

Population Size: 4

Max Iterations: 2

Iteration 1

First step (initialize random population):

Whale	Initial X_1	Initial X_2	$F(x)$
Whale 1	0.54	0.23	3.0243
Whale 2	0.34	0.17	2.0893 (min)
Whale 3	0.84	0.64	6.7788
Whale 4	0.51	0.26	3.1707

Best Solution $X_1=0.34$, $X_2=0.17$, $F(x) = 2.0893$

Update Parameters for $t = 0$

- $a = 2 - (t / \text{Max Iteration}) = 2 - (0 / 2) = 2$
- $a' = -1 + (t / \text{Max Iteration}) = -1 + 0 = -1$
- $A = (2 a r) - a = (2 * 2 * 0.6) - 2 = 0.4$
- $C = 2r = 2 * 0.31 = 0.62$
- $b = (r (a' - L)) + L = (0.23 (-1 - 0.8)) + 0.8 = 0.386$
- $P = 0.12$

r is random num $[0,1]$

L is random num $[0,1]$

P is random num $[0,1]$



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While $P < 0.5$ and $A < 1$ Then update the position

Update Population Positions :

$$D_1 = | (C * \text{Best Solution}) - \text{Current Position} | = | (0.62 * 0.34) - 0.54 | = 0.3292$$

$$D_1 = | (C * \text{Best Solution}) - \text{Current Position} | = | (0.62 * 0.17) - 0.32 | = 0.1246$$

$$\text{New } X_1 = \text{Best Solution} - (A * D) = 0.34 - (0.4 * 0.3292) = 0.20832$$

$$\text{New } X_1 = \text{Best Solution} - (A * D) = 0.17 - (0.4 * 0.1246) = 0.120816$$

$$D_2 = | (0.62 * 0.34) - 0.34 | = 0.1292$$

$$D_2 = | (0.62 * 0.17) - 0.17 | = 0.0646$$

$$\text{New } X_2 = 0.34 - (0.4 * 0.1292) = 0.29832$$

$$\text{New } X_2 = 0.17 - (0.4 * 0.0646) = 0.14416$$

$$D_3 = | (0.62 * 0.34) - 0.84 | = 0.6292$$

$$D_3 = | (0.62 * 0.17) - 0.64 | = 0.5346$$

$$\text{New } X_3 = 0.34 - (0.4 * 0.6292) = 0.08832$$

$$\text{New } X_3 = 0.17 - (0.4 * 0.5346) = -0.04384 = 0$$

$$D_4 = | (0.62 * 0.34) - 0.84 | = 0.2992$$

$$D_4 = | (0.62 * 0.17) - 0.64 | = 0.1546$$

$$\text{New } X_4 = 0.34 - (0.4 * 0.2992) = 0.22032$$

$$\text{New } X_4 = 0.17 - (0.4 * 0.1546) = 0.10816$$

Whale	Initial X_1	Initial X_2	F(x)
Whale 1	0.20832	0.12016	1.3978



Whale Optimization Algorithm (WOA)



Whale 2	0.29832	0.14416	1.796
Whale 3	0.08832	0	0.218 (min)
Whale 4	0.22032	0.10816	1.34

Best Solution $X_1 = 0.08832$, $X_2 = 0$, $F(x) = 0.218$

Iteration 2

- $a = 2 - (1 (2/2)) = 1$
- $a' = -1 + (1 (-1/2)) = -1.5$
- $A = (2 * 1 * 0.99) - 1 = 0.98$
- $C = 2 * 0.2 = 0.4$
- $b = (0.3 (-1.5 - 0.7)) + 0.7 = 0.04$
- $P = 0.6$

While $P > 0.5$ Then update the position

Update Population Positions :

$$D_1 = | \text{Best Solution} - X | = | 0.08832 - 0.20832 | = 0.12$$

$$D_1 = | \text{Best Solution} - X | = | 0 - 0.12016 | = 0.12016$$

$$\text{New } X_1 = D * e^{bl} * \cos(2\pi L) + \text{Best Solution} = 0.21136 \quad L = 0.7$$

$$\text{New } X_1 = D * * \cos(2L) + \text{Best Solution} = 0.12320$$

$$D_2 = | 0.08832 - 0.29832 | = 0.21$$

$$D_2 = | 0 - 0.14416 | = 0.14416$$

$$\text{New } X_2 = D * e^{bl} * \cos(2\pi L) + \text{Best Solution} = 0.30365$$

$$\text{New } X_2 = D * e^{bl} * \cos(2\pi L) + \text{Best Solution} = 0.14781$$

$$D_3 = | 0.08832 - 0.08832 | = 0$$

$$D_3 = | 0 - 0 | = 0$$

$$\text{New } X_3 = D * e^{bl} * \cos(2\pi L) + \text{Best Solution} = 0.08832$$

$$\text{New } X_3 = D * e^{bl} * \cos(2\pi L) + \text{Best Solution} = 0$$

$$D_4 = | 0.08832 - 0.22032 | = 0.132$$



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$$D_4 = |0 - 0.10816| = 0.10816$$

$$\text{New } X_4 = D * e^{bl} * \cos(2\pi L) + \text{Best Solution} = 0.22367$$

$$\text{New } X_4 = D * e^{bl} * \cos(2\pi L) + \text{Best Solution} = 0.11090$$

Whale	Initial X ₁	Initial X ₂	F(x)
Whale 1	0.21136	0.12320	1.42758
Whale 2	0.30365	0.14781	1.83642
Whale 3	0.08832	0	0.218 (min)
Whale 4	0.22367	0.11090	1.36758

Best Solution $X_1 = 0.08832$, $X_2 = 0$, $F(x) = 0.218$

The end of iteration 2.



Chapter 5: Comparison & Conclusion

5.1 Comparison between Whale Optimization Algorithm (WOA) and Ant Colony Optimization (ACO)

Meta-heuristic algorithms are widely used to solve complex optimization problems. Among these algorithms, the Whale Optimization Algorithm (WOA) and Ant Colony Optimization (ACO) are both nature-inspired techniques, but they differ significantly in their mechanisms, performance, and application domains.

1. Inspiration and Basic Concept

The Whale Optimization Algorithm (WOA) is inspired by the bubble-net hunting strategy of humpback whales, where whales encircle their prey and follow a spiral-shaped movement to capture it.

On the other hand, Ant Colony Optimization (ACO) is inspired by the foraging behavior of ants, where ants deposit pheromones on paths to communicate and guide other ants toward the shortest route.

2. Search Mechanism

WOA mainly relies on three mechanisms:

- Encircling prey
- Bubble-net attacking (exploitation)
- Search for prey (exploration)

These mechanisms allow WOA to dynamically switch between exploration and exploitation.

In contrast, ACO depends on:

- Pheromone updating
- Probabilistic path selection
- Evaporation mechanism

ACO builds solutions step-by-step based on probability and collective learning from previous iterations.



3. Exploration vs Exploitation

WOA achieves a balance between exploration and exploitation using adaptive parameters (such as coefficient A):

- When $|A| > 1 \rightarrow$ exploration
- When $|A| < 1 \rightarrow$ exploitation

ACO, however, emphasizes exploitation through pheromone reinforcement, which may lead to faster convergence but increases the risk of premature convergence if diversity is not maintained.

4. Problem Type Suitability

WOA is more suitable for:

- Continuous optimization problems
- Engineering design
- Parameter tuning

ACO is more effective for:

- Discrete optimization problems
- Combinatorial problems
- Routing problems (e.g., Traveling Salesman Problem)

5. Computational Complexity

WOA has:

- Lower computational complexity
- Fewer parameters to tune
- Simpler implementation

ACO has:

- Higher computational cost due to pheromone updates
- More parameters (pheromone importance, evaporation rate, etc.)
- More complex implementation

6. Convergence Behavior

WOA:



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- Provides stable convergence
- May be slower in some cases
- Can avoid local optima using random search behavior

ACO:

- Converges faster in structured problems
- May get trapped in local optima due to pheromone accumulation

7. Performance and Accuracy

Studies show that:

- WOA performs well in continuous benchmark functions
- ACO performs better in discrete path-finding problems

Thus, performance depends highly on the nature of the problem.

Comparison Table

Feature	WOA	ACO
Inspiration	Whale hunting	Ant foraging
Problem Type	Continuous	Discrete
Complexity	Low	High
Parameters	Few	Many
Convergence	Stable	Fast but risky
Exploration	Strong	Limited
Best Application	Engineering optimization	Routing & graph problems

5.2 Conclusion

The Whale Optimization Algorithm (WOA) has emerged as a powerful and efficient meta-heuristic optimization technique, demonstrating strong performance across a wide range of complex optimization problems. Inspired by the bubble-net hunting behavior of humpback whales, WOA successfully translates a natural predatory strategy into a mathematical framework capable of solving nonlinear, non-convex, and high-dimensional problems.

Based on the analysis presented throughout this report, WOA offers several key advantages, including its conceptual simplicity, minimal number of control parameters, and effective balance between exploration and exploitation. These characteristics enable the algorithm to achieve



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competitive solution accuracy and stable convergence behavior, particularly in continuous optimization domains such as engineering design, machine learning, and robotics.

In comparison with Ant Colony Optimization (ACO), clear distinctions can be observed. While ACO excels in discrete and combinatorial problems—especially routing and graph-based optimization due to its pheromone-guided search mechanism—WOA proves to be more suitable for continuous search spaces. Additionally, WOA requires less computational effort and is easier to implement, whereas ACO involves more complex parameter tuning and higher computational overhead.

Despite its strengths, WOA is not without limitations. It may experience slower convergence in certain scenarios and can occasionally become trapped in local optima, particularly in highly complex search spaces. However, recent research has addressed these challenges through various enhancements such as hybridization techniques, adaptive parameter control, and chaotic initialization strategies, which significantly improve its overall performance.

In conclusion, the choice between WOA and ACO is highly dependent on the nature of the problem. WOA is generally more effective for continuous optimization tasks, while ACO remains a preferred approach for discrete and combinatorial problems. Future research directions may focus on hybrid models that integrate the strengths of both algorithms, aiming to achieve improved robustness, faster convergence, and higher solution quality across diverse application domains.



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